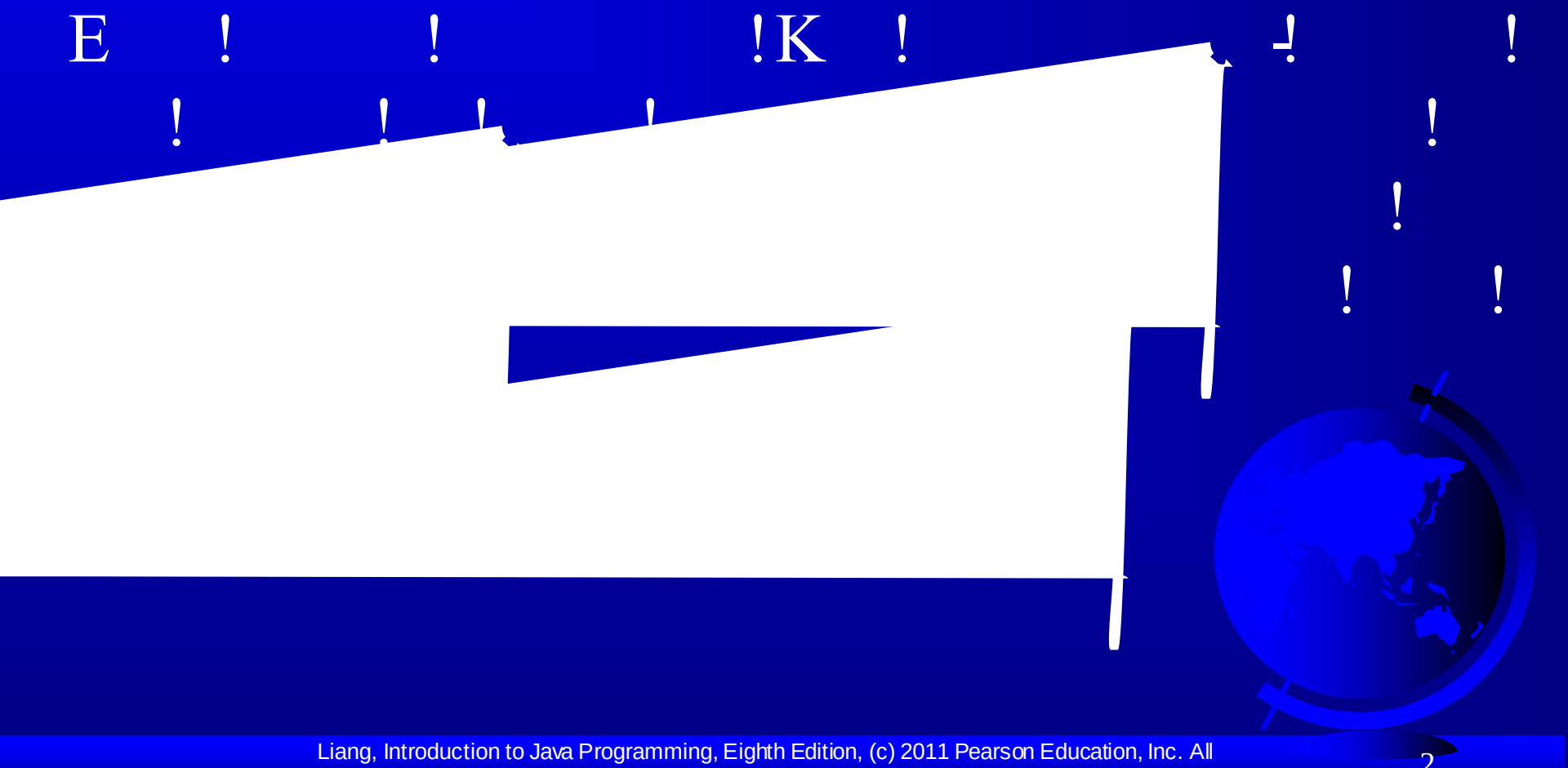


C ! 4;! (Selections)

Motivasyon



B

- boolean tipi ve ! ! Boolean ! ! (§3.2).
- C ! ! ! AdditionQuiz ! ! (§3.3).
- ! ! - ! ! ! ! ! (§3.4)
- ! ! ! ! !!GuessBirthday oyunu yazma (§3.5).
- ! ! - ! ! ! ! (§3.6).
- ! ! ! !) *! ! ! ! (§3.7).
- if ! ! ! ! (§3.8).
- ! !(BMI, ComputeTax, SubtractionQuiz) (§3.9-3.11).
- N !) *! ! ! ! (&&, ||, and !) (§3.12).
- ! ! ! ! ! ! (LeapYear, Lottery) (§§3.13-3.14).
- switch ! ! ! (§3.15).
- L ! ! ! (§3.16).
- System.out.printf met ! ! ! (§3.17).
- P ! ! ! ! ! ! ! (§3.18).

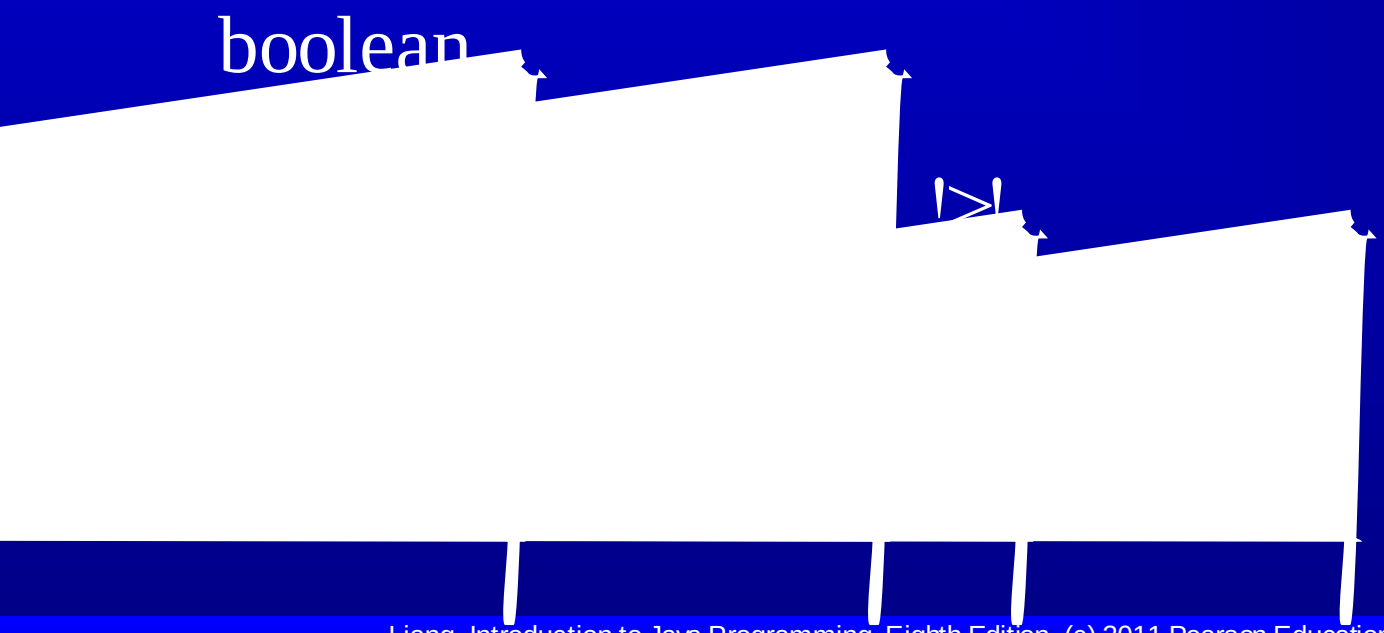
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`boolean` veri tipi



`boolean` veri tipi

İlkel tip	Sınıf	Uzunluk (byte)	Kapsam
boolean	boolean	1	TRUE FALSE



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L !P

Operatör	*Ad*
<	
<=	! !
>	
>=	! !
==	
!=	!



Q ;!C !N ! !B !

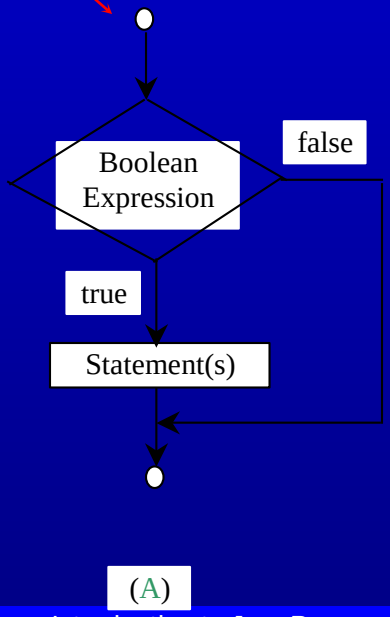
Bu 1. toplama egzersizleri bir program isteniyor. Program rastgele tek 1 ve 2 gibi iki tane tam ve 7 + 9 gibi bu iki sorar. sonra, program belirten bir mesaj verir.

AdditionQuiz

Run

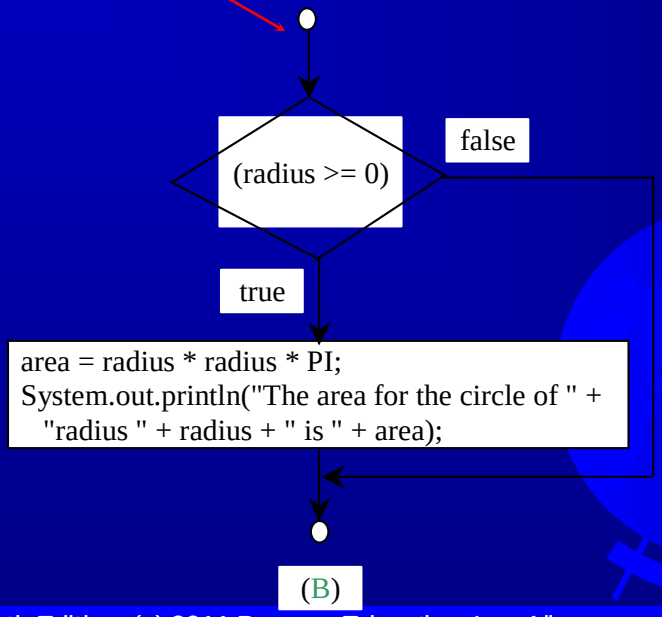
Tek-if ifadeleri

```
if (boolean-expression) {
  statement(s);
}
```



(A)

```
if (radius >= 0) {
  area = radius * radius * PI;
  System.out.println("The area"
    + " for the circle of radius "
    + radius + " is " + area);
}
```



(B)

Not

```
if i > 0 {
  System.out.println("i is positive");
}
```

(a)

```
if (i > 0) {
  System.out.println("i is positive");
}
```

(b) E

```
if (i > 0) {
  System.out.println("i is positive");
}
```

(a)

Denktir

```
if (i > 0)
  System.out.println("i is positive");
```

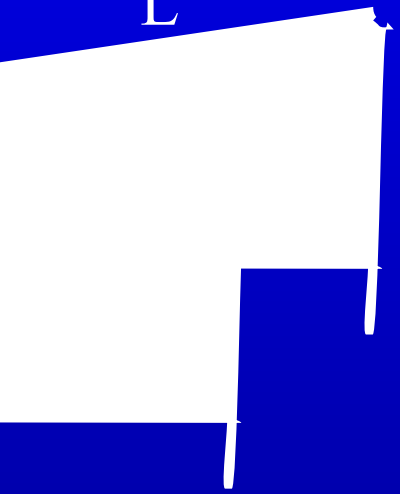
(b)



C ! !

L ! ! ! ! ! ! ! !

! 6(! -! HiFive -!3! ! ! HiEven

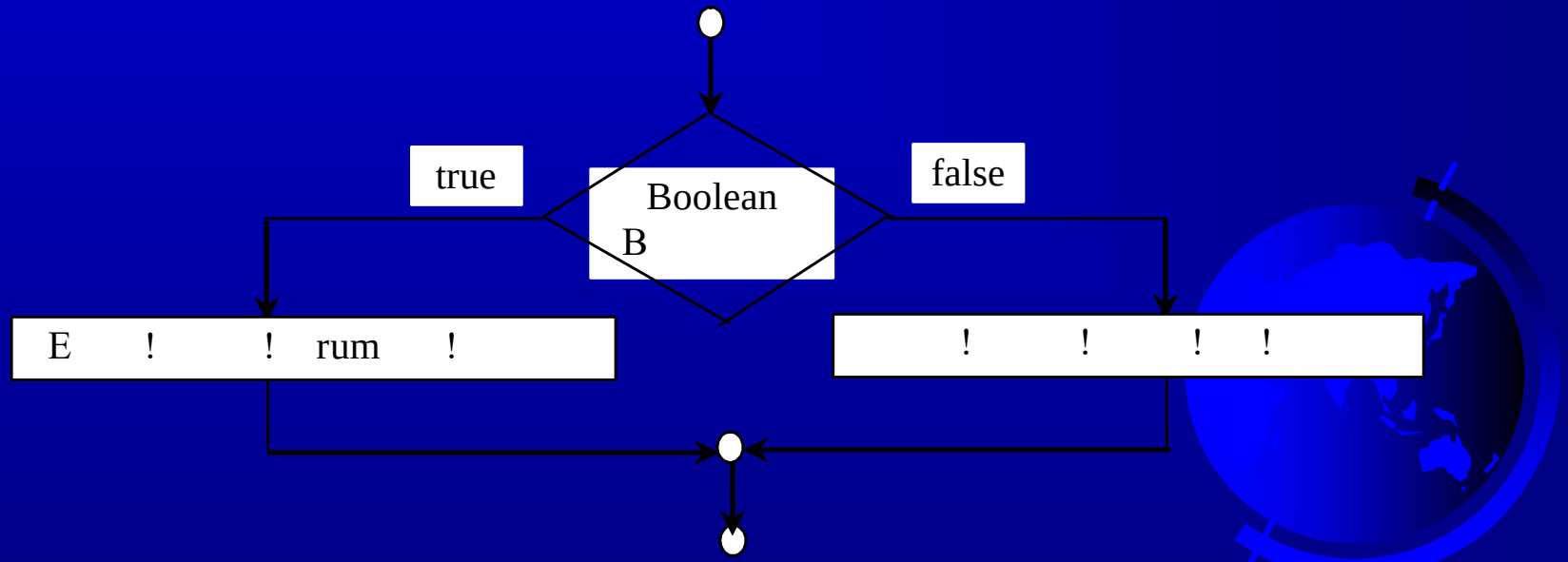


SimpleIfDemo

Run

- if ifadesi

```
if (boolean-expression) {
  statement(s)-for-the-true-case;
}
else {
  statement(s)-for-the-false-case;
}
```




Learning

! !) *!if ifadeleri

```
if (score >= 90.0)
  grade = 'A';
else
  if (score >= 80.0)
    grade = 'B';
  else
    if (score >= 70.0)
      grade = 'C';
    else
      if (score >= 60.0)
        grade = 'D';
      else
        grade = 'F';
```

Equivalent

```
if (score >= 90.0)
  grade = 'A';
else if (score >= 80.0)
  grade = 'B';
else if (score >= 70.0)
  grade = 'C';
else if (score >= 60.0)
  grade = 'D';
else
  grade = 'F';
```

if-else !

Skorun 70.0 !

L ! FALSE olur

```
if (score >= 90.0)
  grade = 'A';
else if (score >= 80.0)
  grade = 'B';
else if (score >= 70.0)
  grade = 'C';
else if (score >= 60.0)
  grade = 'D';
else
  grade = 'F';
```



if-else

Skorun 70.0 !

L !GBM F!

```
if (score >= 90.0)
  grade = 'A';
else if (score >= 80.0)
  grade = 'B';
else if (score >= 70.0)
  grade = 'C';
else if (score >= 60.0)
  grade = 'D';
else
  grade = 'F';
```



if-else

!

Skorun 70.0 !

L ! F!

```
if (score >= 90.0)
  grade = 'A';
else if (score >= 80.0)
  grade = 'B';
else if (score >= 70.0)
  grade = 'C';
else if (score >= 60.0)
  grade = 'D';
else
  grade = 'F';
```



if-else

!

Skorun 70.0 !

Harf notu C olur

```
if (score >= 90.0)
  grade = 'A';
else if (score >= 80.0)
  grade = 'B';
else if (score >= 70.0)
  grade = 'C';
else if (score >= 60.0)
  grade = 'D';
else
  grade = 'F';
```



if-else

!

Skorun 70.0 !

J ! !

```
if (score >= 90.0)
  grade = 'A';
else if (score >= 80.0)
  grade = 'B';
else if (score >= 70.0)
  grade = 'C';
else if (score >= 60.0)
  grade = 'D';
else
  grade = 'F';
```



Not

F ! ! ! ! ! ! ! !

```
int i = 1;
int j = 2;
int k = 3;

if (i > j)
  if (i > k)
    System.out.println("A");
else
    System.out.println("B");
```

(a)

Equivalent

```
int i = 1;
int j = 2;
int k = 3;

if (i > j)
  if (i > k)
    System.out.println("A");
  else
    System.out.println("B");
```

(b)




Hang

!

! ;! ! ! ! !

```
if (radius >= 0);
{
  area = radius*radius*PI;
  System.out.println(
    "The area for the circle of radius " +
    radius + " is " + area);
}
```

Wrong

C ! ! ! ! ! ! ! ! !

C ! !

```
if (number % 2 == 0)
  even = true;
else
  even = false;
```

(a)

Denktir

```
boolean even
  = number % 2 == 0;
```

(b)



Dikkat

```
if (even == true)
  System.out.println(
    "It is even.");
```

(a)

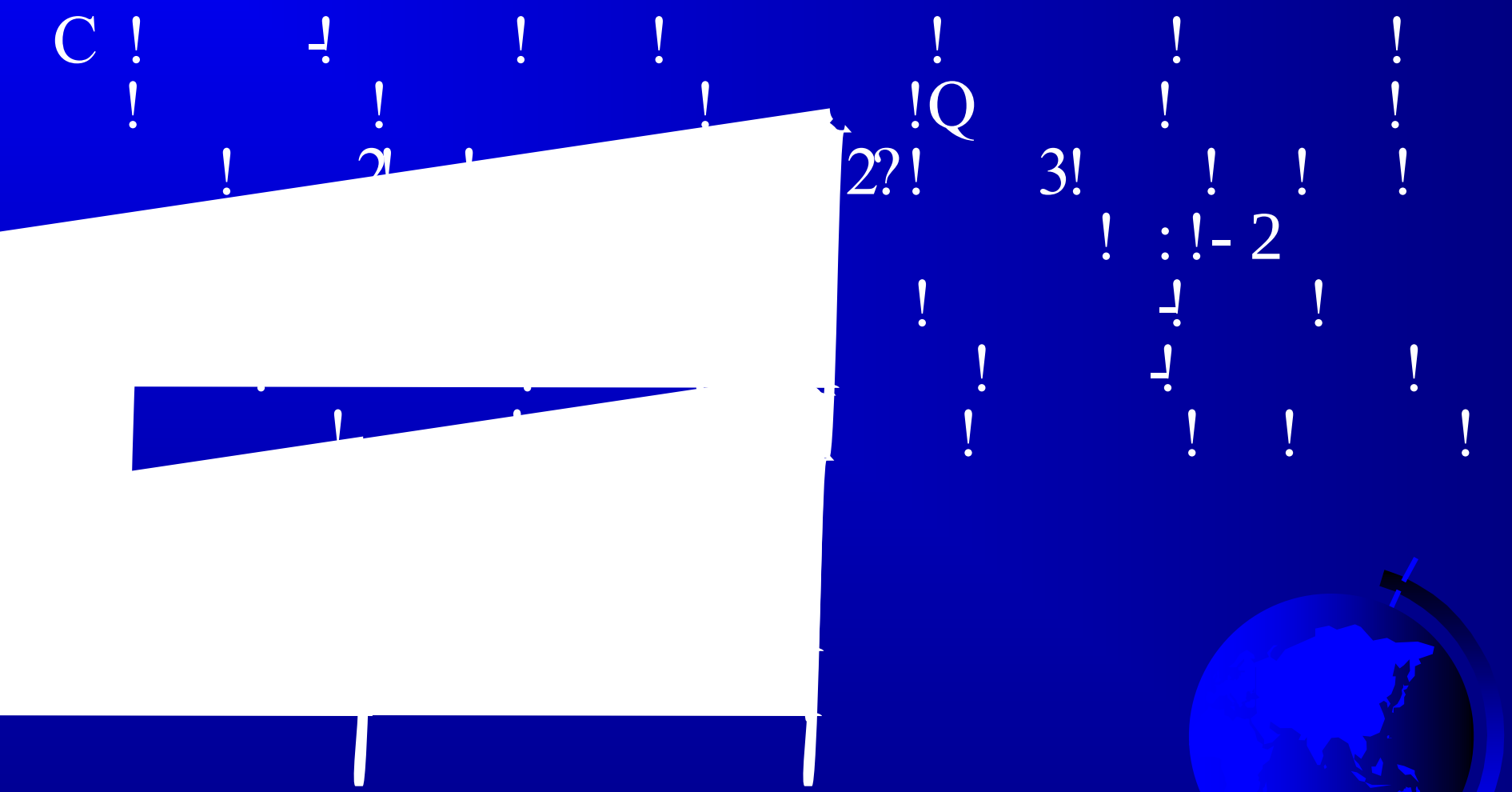
Denktir

```
if (even)
  System.out.println(
    "It is even.");
```

(b)



Problem: H !N ! !B



SubtractionQuiz

Run

Problem: !L !

!L ! !)Body Mass Index (BMI))
! ! !L !
metre karesine
! 27! ! ! !
:

BMI	Yorum
16 altı	çok zayıf
16-18	zayıf
18-24	normal
24-29	biraz kilolu
29-35	aşırı kilolu
35 üzeri	obez

ComputeBMI

Run

Problem: Vergi Hesaplama

BCE! ! ! ! ! ! !

inal Rate	Single	Married Filing Jointly or Qualified Widow(er)	Married Filing Separately	Head of Household
10%	$0 $8,350	$0 $16,700	$0 $8,350	$0 $11,950
15%	$8,351 $33,950	$16,701 $67,900	$8,351 $33,950	$11,951 $45,500
25%	$33,951 $82,250	$67,901 $137,050	$33,951 $68,525	$45,501 $117,450
28%	$82,251 $171,550	$137,051 $208,850	$68,525 $104,425	$117,451 $190,200
33%	$171,551 $372,950	$208,851 $372,950	$104,426 $186,475	$190,201 - $372,950
35%	$372,951+	$372,951+	$186,476+	$372,951+

Problem:Vergi Hesaplama, devam

```
if (status == 0) {
  // Compute tax for single filers
}
else if (status == 1) {
  // Compute tax for married file jointly
}
else if (status == 2) {
  // Compute tax for married file separately
}
else if (status == 3) {
  // Compute tax for head of household
}
else {
  // Display wrong status
}
```



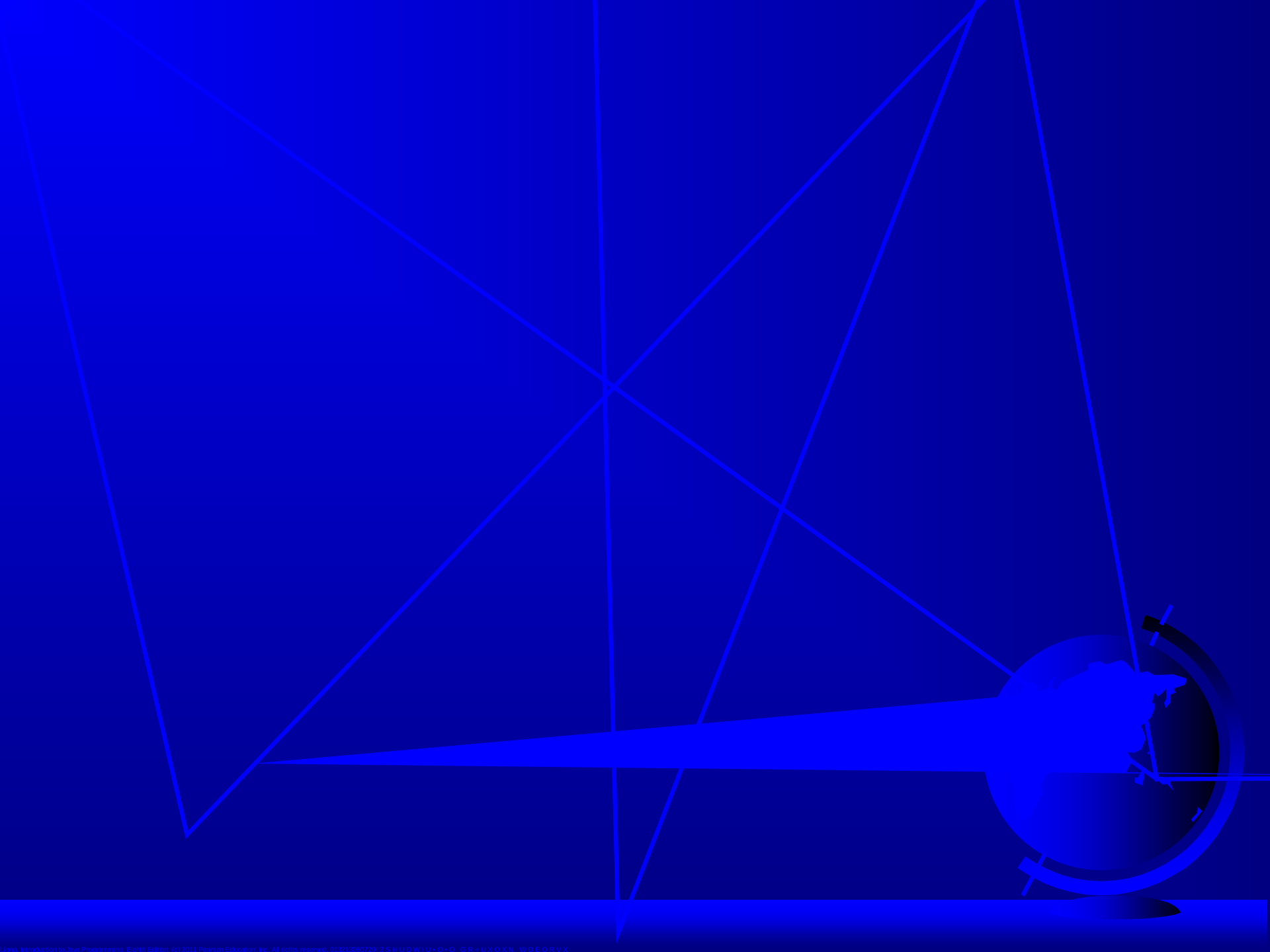
ComputeTax

Run

N !P !
(Logical Operators)

Operatör	*Ad*
!	not
& &	and
| |	or
^	exclusive or





Liang, Introduction to Java Programming, Eighth Edition, (c) 2011 Pearson Education, Inc. All rights reserved. 0132130807 2 3 4 5 6 7 8 9 10 V W U X + Q = P L G F = U X O X N W D E F U B Y %

|| P !E !

p1	p2	p1 || p2
false	false	false
false	true	true
true	false	true
true	true	true

Example (assume age = 24, gender = 'F')

(age > 34) || (gender == 'F') is true, because (gender == 'F') is true.

(age > 34) || (gender == 'M') is false, because (age > 34) and (gender == 'M') are both false.

p1	p2	p1 || p2
false	false	false
false	true	true
true	false	true
true	true	true

Example

(2 > 3) || (5 > 5) is false, because (2 > 3) and (5 > 5) are both false.

(3 > 2) || (5 > 5) is true, because (3 > 2) is true.

^ P !E !

p1	p2	p1 ^ p2
false	false	false
false	true	true
true	false	true
true	true	false

Example (assume age = 24, gender = 'F')

(age > 34) ^ (gender == 'F') is true, because (age > 34) is false but (gender == 'F') is true.

(age > 34) || (gender == 'M') is false, because (age > 34) and (gender == 'M') are both false.





Companion Website

& ve | O

```
Eğer x, 1 ise bu açıklamadan sonra x
  ne olur?
(x > 1) & (x++ < 10)

Eğer x, 1 ise bu açıklamadan sonra x
  ne olur?
(1 > x) && ( 1 > x++)

(1 == x) | (10 > x++)?
(1 == x) || (10 > x++)?
```



Problem:B ! ! ?

C ! ! !

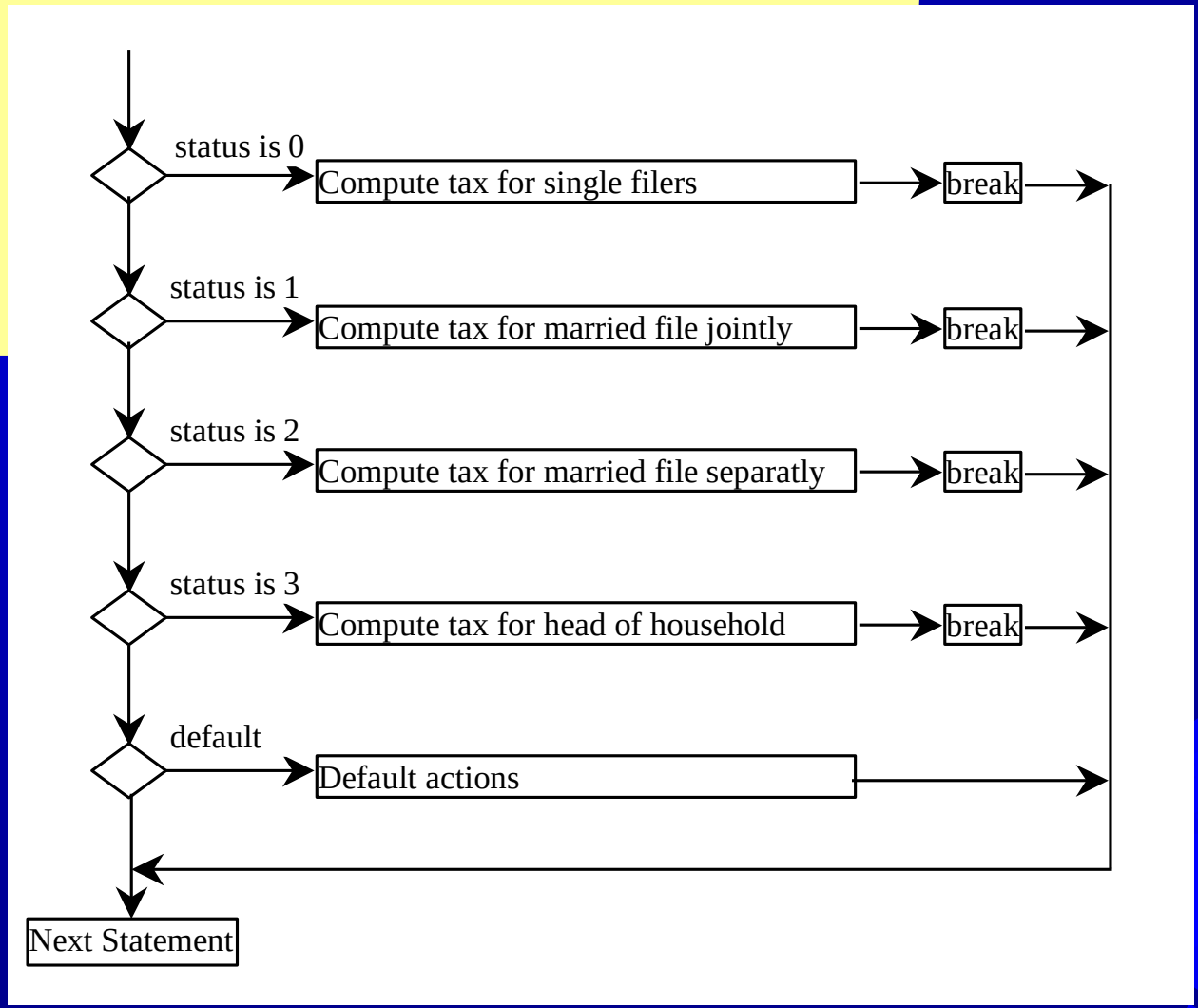


switch

```
switch (durum) {
  case 0:      !              !!     !vergi hesapla;
         break;
  case 1: ortak evli          !!     !vergi hesapla ;
         break;
  case 2:      !     !        !!     !vergi hesapla ;
         break;
  case 3: hane reisi          !!     !vergi hesapla ;
         break;
  default: System.out.println("  I     ;!          !        ");
         System.exit(0);
}
```



Switch


status is 0
Compute tax for single filers
break
status is 1
Compute tax for married file jointly
break
status is 2
Compute tax for married file separatly
break
status is 3
Compute tax for head of household
break
default
Default actions
Next Statement

switch !

Switch ifadesi char, bayt, short
! ! ! ! !
vermeli ve her zaman parantez
!

value1, ... ve valueN, anahtar
! ! ! ! !
! !D ! !
! ! ! !
-! ! ! !
! ! 2! ! !
O(! ! ! -!
!2, ! ! ! !
! !

```
switch (switch-expression) {
  case value1:  statement(s)1;
                break;
  case value2: statement(s)2;
                break;

  case valueN: statement(s)N;
                break;
  default: statement(s)-for-default;
}
```



switch !

B ! ! ! !
-! ! ! !
sonunda switch ifadesinin
! ! !
!C ! !
yoksa, bir sonraki durum
!

```
switch (switch-expression) {
case value1:  statement(s)1;
              break;
case value2: statement(s)2;
              break;

case valueN: statement(s)N;
              break;
default: statement(s)-for-default;
}
```

! ! ! -!
! ! !
! ! !
! ! !

E !) *! ! ! -!
! ! !) ! !
*! ! !C ! -!
! ! ! ! ! !
! ! ! ! ! !

bir programlama stilidir.



switch !

ch -(olsun:

```
switch (ch) {
  case 'a': System.out.println(ch);
  case 'b': System.out.println(ch);
  case 'c': System.out.println(ch);
}
```



switch !

!(! :

```
switch (ch) {
  case 'a': System.out.println(ch);
  case 'b': System.out.println(ch);
  case 'c': System.out.println(ch);
}
```



switch !

C ! !

```
switch (ch) {
  case 'a': System.out.println(ch);
  case 'b': System.out.println(ch);
  case 'c': System.out.println(ch);
}
```



switch !

C ! !

```
switch (ch) {
  case 'a': System.out.println(ch);
  case 'b': System.out.println(ch);
  case 'c': System.out.println(ch);
}
```



switch !

C ! !

```
switch (ch) {
  case 'a': System.out.println(ch);
  case 'b': System.out.println(ch);
  case 'c': System.out.println(ch);
}
```



switch !

H ! !

```
switch (ch)
  case 'a': System.out.println(ch);
  case 'b': System.out.println(ch);
  case 'c': System.out.println(ch);
}
```

Next statement;



switch !

ch (olsun:

```
switch (ch) {
  case 'a': System.out.println(ch);
            break;
  case 'b': System.out.println(ch);
            break;
  case 'c': System.out.println(ch);
}
```



switch !

!(! :

```
switch (ch) {
  case 'a': System.out.println(ch);
            break;
  case 'b': System.out.println(ch);
            break;
  case 'c': System.out.println(ch);
}
```



switch !

C ! !

```
switch (ch) {
  case 'a': System.out.println(ch);
            break;
  case 'b': System.out.println(ch);
            break;
  case 'c': System.out.println(ch);
}
```

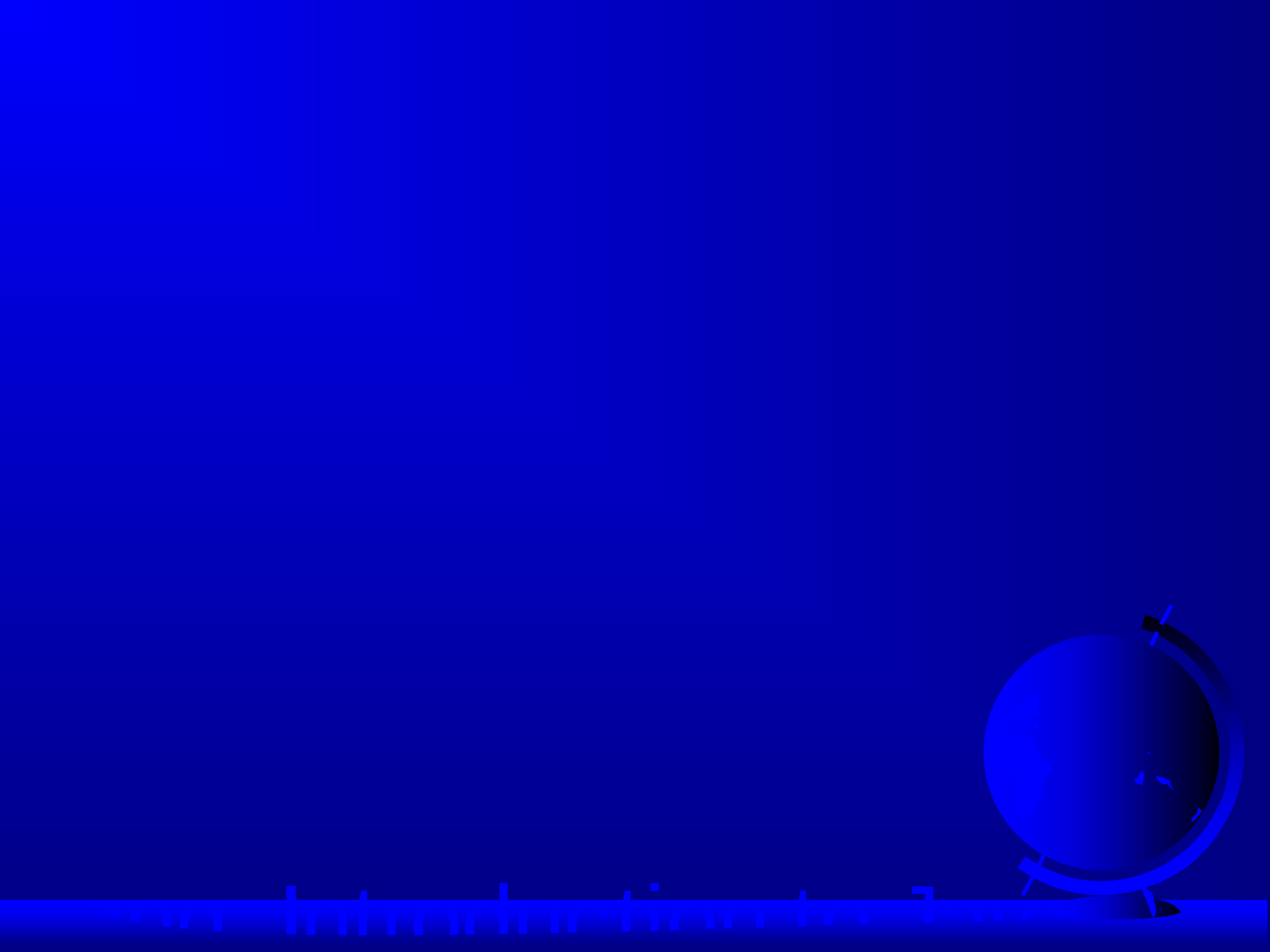


switch !

C ! !

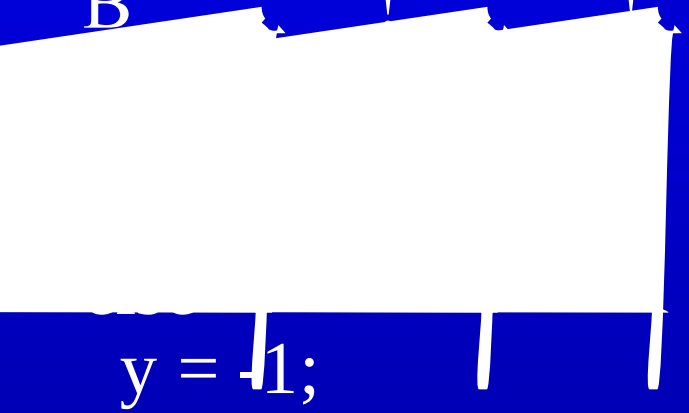
```
switch (ch) {
  case 'a': System.out.println(ch);
            break;
  case 'b': System.out.println(ch);
            break;
  case 'c': System.out.println(ch);
}
```





L !P (Conditional Operator)

B ;



y = -1;

y = (x > 0) ? 1 : -1;

(boolean-expression) ? expression1 : expression2

Ternary operator
Binary operator
Unary operator

L !P

```
if (num % 2 == 0)
  System.out.println(num + “is even”);
else
  System.out.println(num + “is odd”);
```

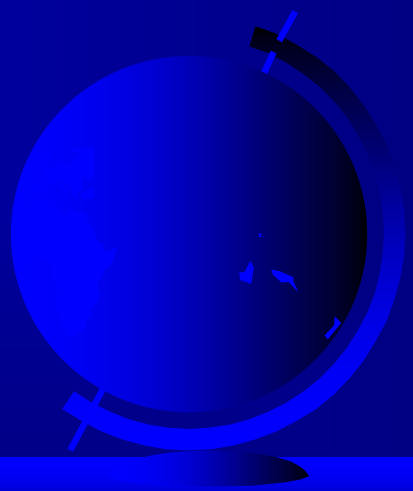
```
System.out.println(
  (num % 2 == 0)? num + “is even” :
  num + “is odd”);
```



L !P

```
(boolean-expression) ? exp1 : exp2
```



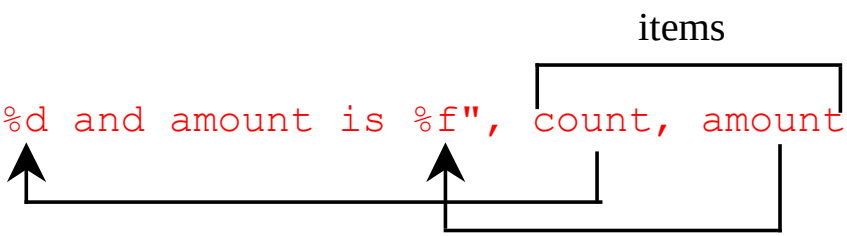


!L !C

Specifier	Output	Example
%b	a boolean value	true or false
%c	a character	'a'
%d	a decimal integer	200
%f	a floating-point number	45.460000
%e	a number in standard scientific notation	4.556000e+01
%s	a string	"Java is cool"

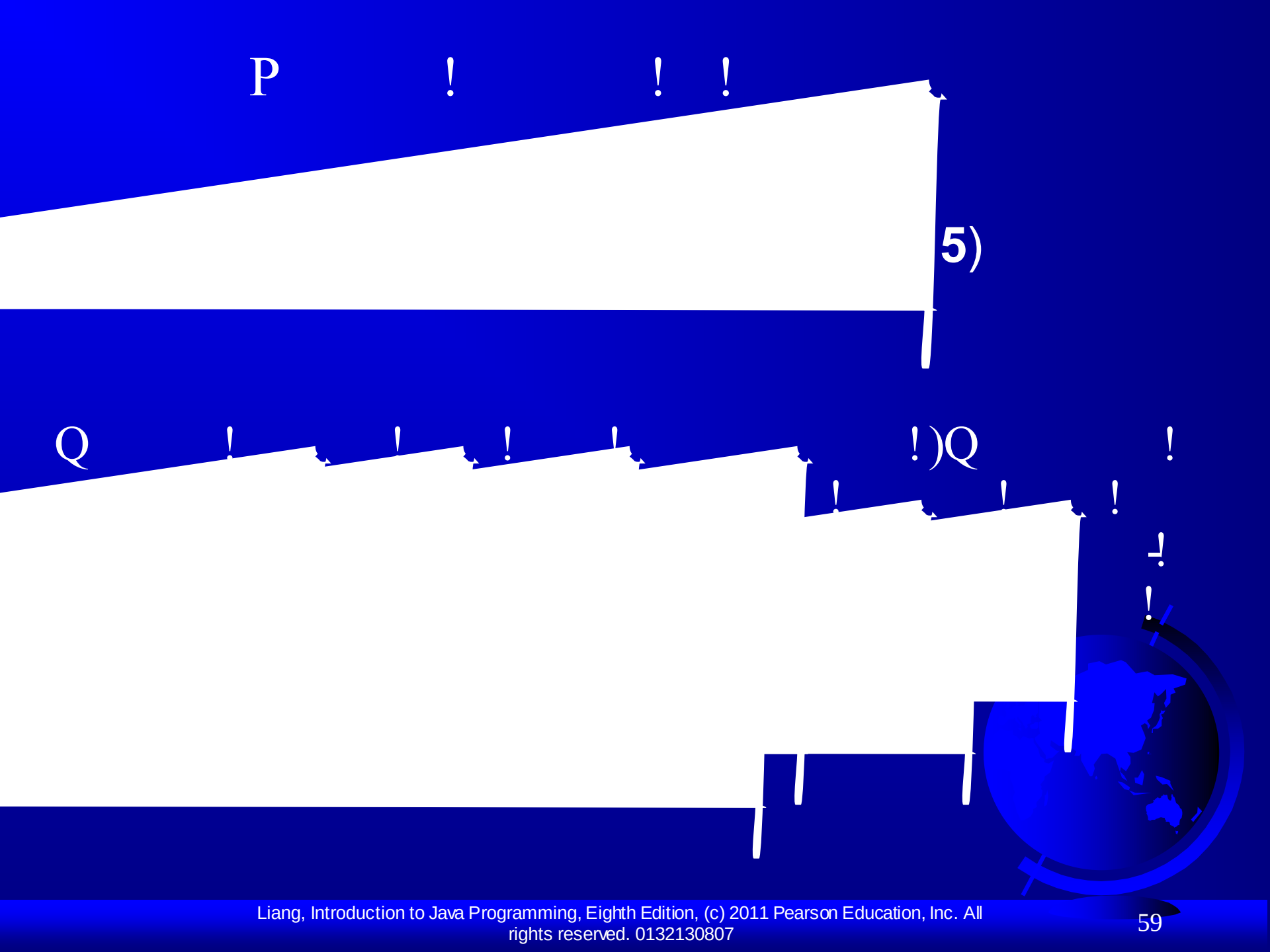
```
int count = 5;
double amount = 45.56;
System.out.printf("count is %d and amount is %f", count, amount);


display              count is 5 and amount is 45.560000
```



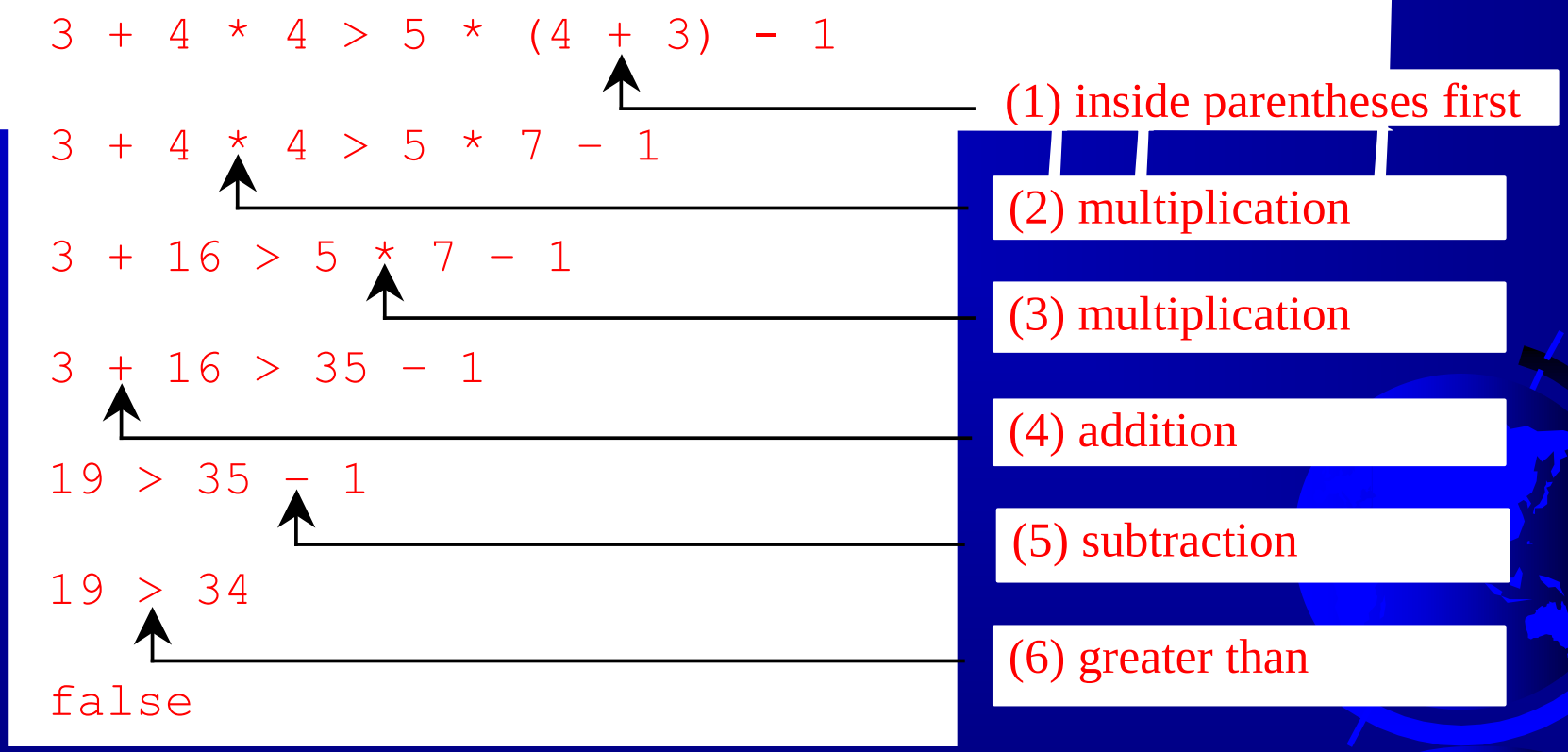
Operat ! !

Precedence	*Operator*
↓	`var++` and `var--` (Postfix)
	`+`, `-` (Unary plus and minus), `++var` and `--var` (Prefix)
	(type) (Casting)
	`!` (Not)
	`*`, `/`, `%` (Multiplication, division, and remainder)
	`+`, `-` (Binary addition and subtraction)
	`<`, `<=`, `>`, `>=` (Relational)
	`==`, `!=` (Equality)
	`^` (Exclusive OR)
	`&&` (AND)
	`||` (OR)
	`=`, `+=`, `-=`, `*=`, `/=`, `%=` (Assignment operators)





P ! ! ! ! ! !4!
+ 4 * 4> 5 *(4 ! ! ! ;


3 + 4 * 4 > 5 * (4 + 3) - 1
(1) inside parentheses first
3 + 4 * 4 > 5 * 7 - 1
(2) multiplication
3 + 16 > 5 * 7 - 1
(3) multiplication
3 + 16 > 35 - 1
(4) addition
19 > 35 - 1
(5) subtraction
19 > 34
(6) greater than
false